

VR Rehabilitation Coach

Unity-based upper-limb bicep curl rehabilitation system

Platform: Unity + Meta Quest

Focus: Bicep curl, feedback, CSV logging

Objectives

- Develop a Unity-based VR rehabilitation application for upper-limb bicep curl training.
- Provide real-time feedback to improve exercise form, control, and consistency.
- Create a system that produces research-ready movement data for lab use.

Technical Approach

- Use Unity with Meta Quest headset/controllers for VR motion tracking.
- Create a clean rehab room with a grabbable dumbbell, instruction panel, rep counter, and score panel.
- Estimate elbow angle, range of motion, pace, rep count, and movement smoothness from controller data.
- Display feedback cues such as "Raise higher," "Lower more slowly," "Good rep," and "Avoid swinging."

Research Directions

- Review literature on VR rehabilitation, controller-based motion tracking, and upper-limb exercise feedback.
- Use bicep curl as the first baseline exercise before expanding to additional movements.
- Evaluate whether Quest controller data can support movement-quality assessment and CSV logging.
- Design the logging schema so Delsys IMU fields can be added later without changing the structure.

Expected Accomplishments

- Deliver a working VR prototype with a grabbable virtual dumbbell and functional bicep curl exercise.
- Include real-time green / yellow / red feedback, rep counting, and a basic movement-quality score.
- Generate CSV output with participant/session data, controller samples, estimated elbow angle, smoothness score, rep success flag, and feedback message.
- Provide documentation and a short demo video, while establishing a foundation for future IMU integration and additional exercises.